# INTRODUCTION TO SOFTWARE ENGINEERING

## Part II – Software Design

Software Engineering Department

College of Information and Communication Technology

# Software Process

- Requirements analysis & specification (Definition)
- Design
- Implementation (Programming)
- Testing
- Delivery and Maintenance

# References

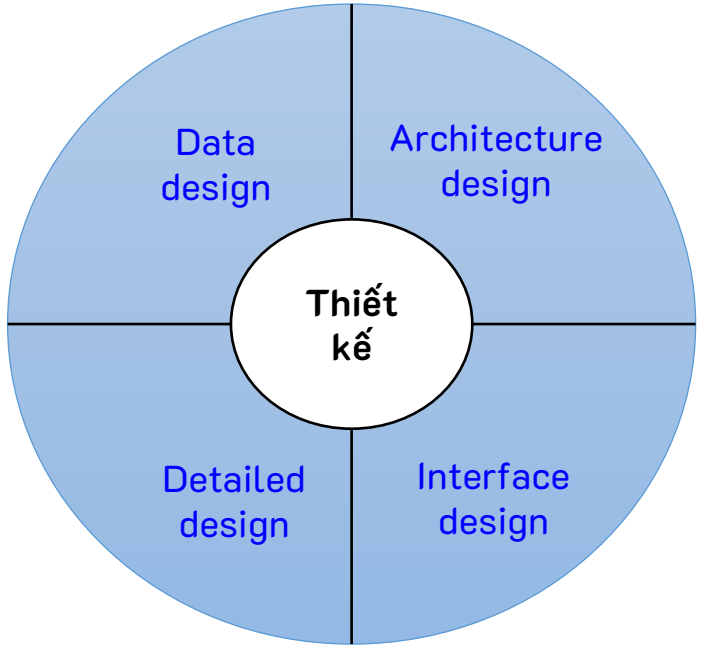

- Shari Lawrence Pfleeger, Joanne M.Atlee, Software Engineering theory and practice, Pearson Prentice Hall, 4th edition, 2010.
- Ian Sommerville, Software Engineering, Addison – Wesley, 9th edition, 2009.
- Roger S. Pressman, Software Engineering: A Practitioner's Approach - 7th edition, McGraw-Hill, 2009.
- Thomas Connolly and Carolyn Begg, Database Systems: A Practical Approach to Design, Implementation, and Management - 6th edition, Pearson Education Limited, 2015.
- Hiep Xuan Huynh, Lan Phuong Phan, Introduction to Software Engineering, Can Tho university publishing house, 2010.

# Design

- Design is the creative process of figuring out how to implement all of the customer's requirements; the resulting plan is also called the design
- Early design decisions address the system's architecture
- Later design decisions address how to implement the individual units

# Design

- Main design levels:
  - **Architectural design:** Decompose the system into large building blocks, choose the tech stack.
  - **Data design:** Organize, store, and protect the integrity of information.
  - **Interface design:** APIs between systems, user interfaces between users and the system (UI/UX).
  - **Detailed design:** Design classes and their internal behaviors.

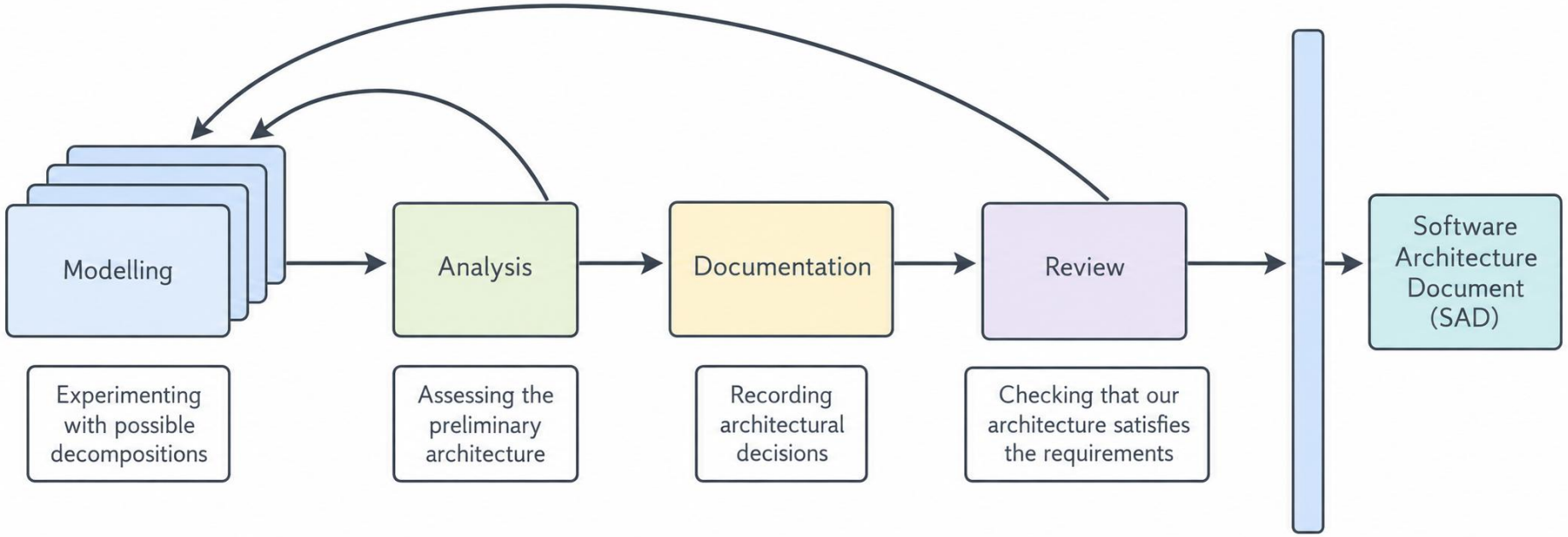


# Architectural Design

- Provides a comprehensive view of the system to be built.
  - Concerns the components, connections, and constraints on the combined components.
  - Links the system components to the capabilities identified in the requirements specification.
  - Software architectures also include general solutions, known as architectural styles.

# Design Process

- Designing software system is an iterative process
- The final outcome is the software architecture document (SAD)

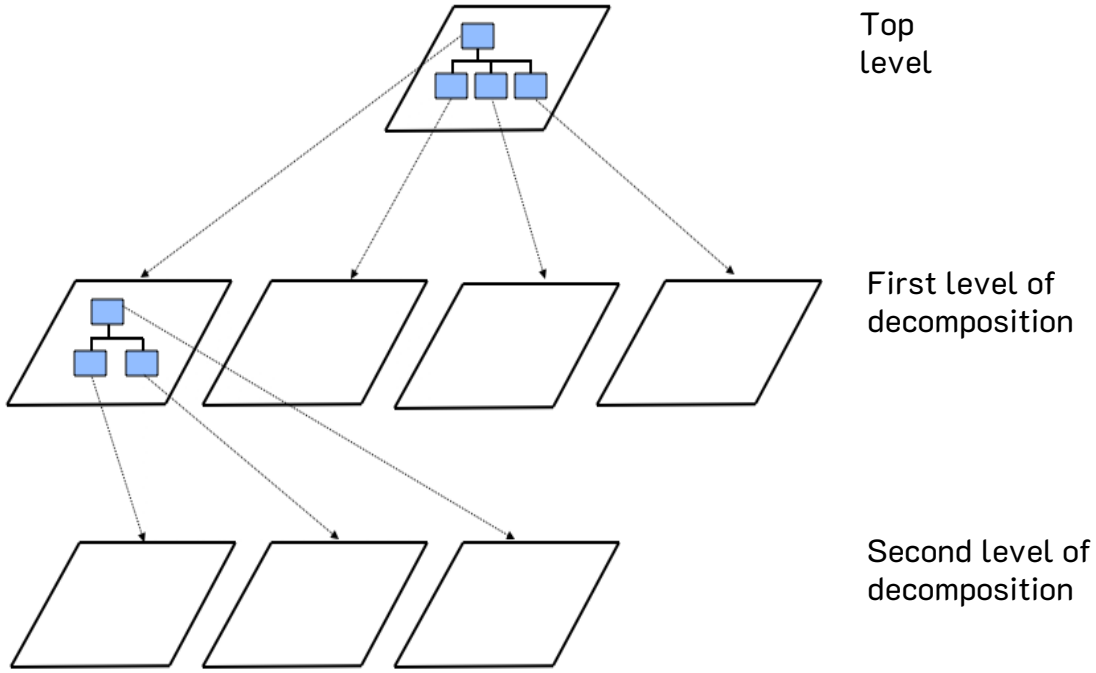


# Modeling Architectures

- Collection of models helps to answer whether the proposed architecture meets the specified requirements
- Six ways to use the architectural models:
  - to understand the system
  - to determine amount of reuse from other systems and the reusability of the system being designed
  - to provide blueprint for system construction
  - to reason about system evolution
  - to analyze dependencies
  - to support management decisions and understand risks

# Decomposition and Views

- High-level description of system's key elements
- Creating a hierarchy of information with increasing details



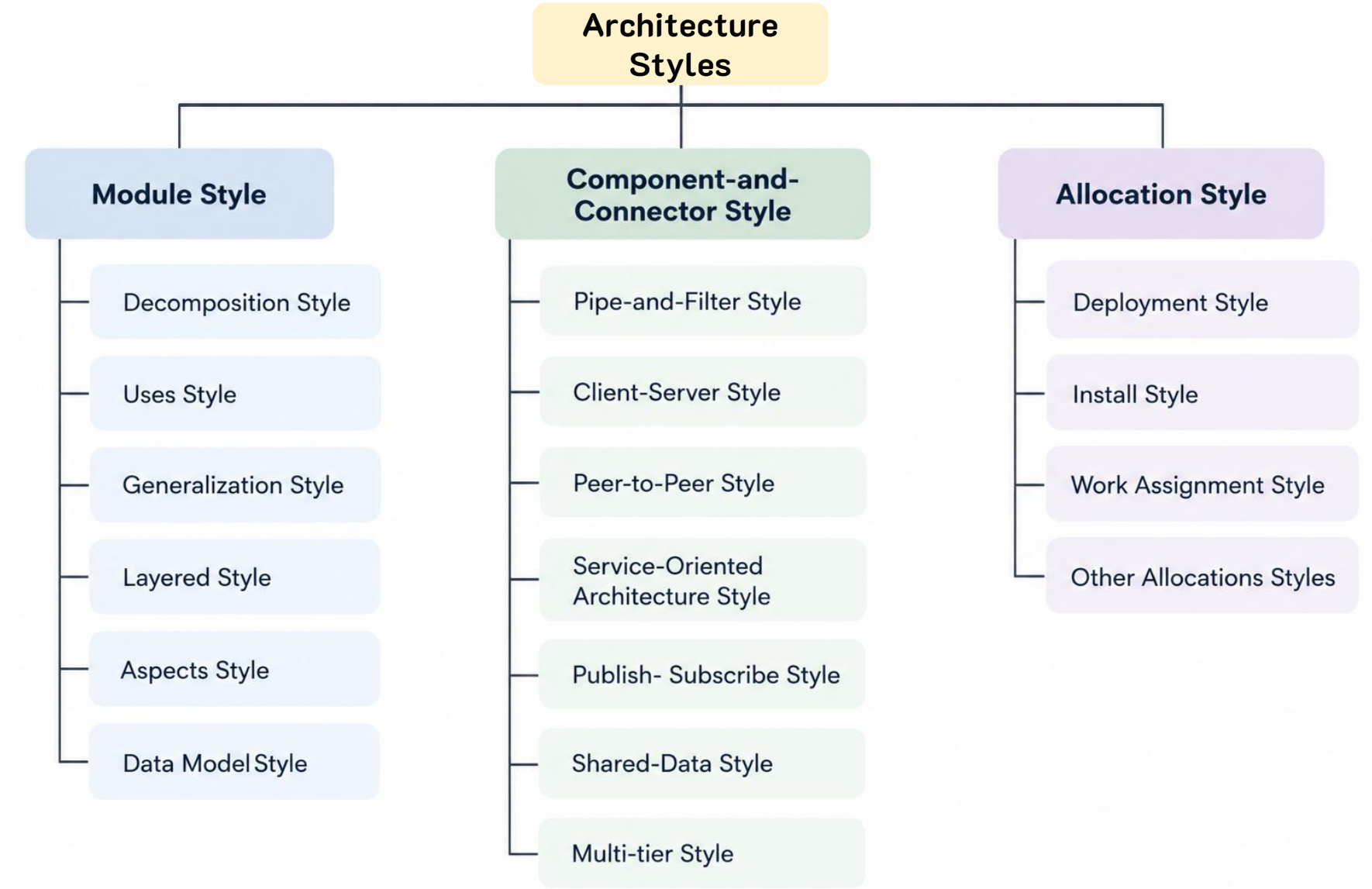
# Decomposition and Views – Popular Design Methods

- Some design problems have no existing solutions
  - Designers must decompose to isolate key problems
- Some popular design methods:
  - Functional decomposition
  - Feature-oriented decomposition
  - Data-oriented decomposition
  - Process-oriented decomposition
  - Event-oriented decomposition
  - Object-oriented design

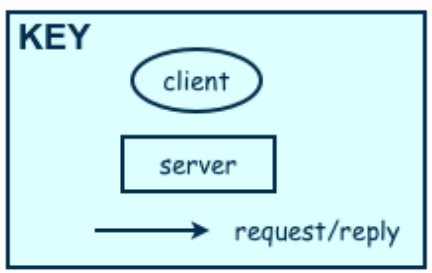
# Decomposition and Views – Architectural Views

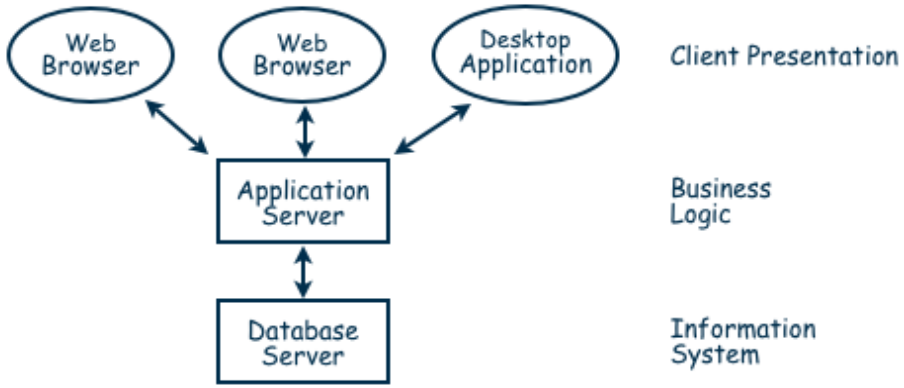
- Common types of architectural views include:
  - Decomposition view
  - Dependencies view
  - Generalization view
  - Execution view
  - Implementation view
  - Deployment view
  - Work-assignment view

# Architectural Styles and Strategies

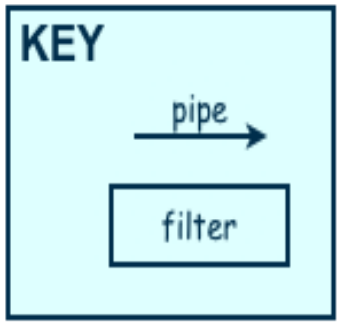


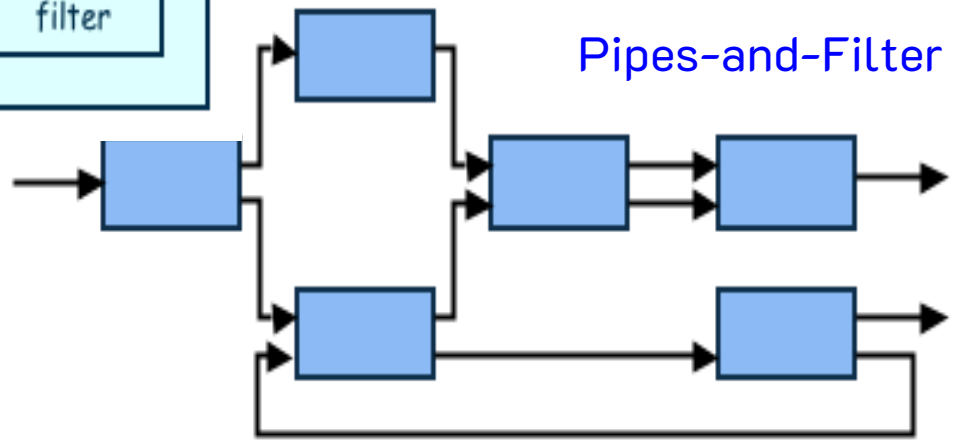
# Architectural Styles and Strategies



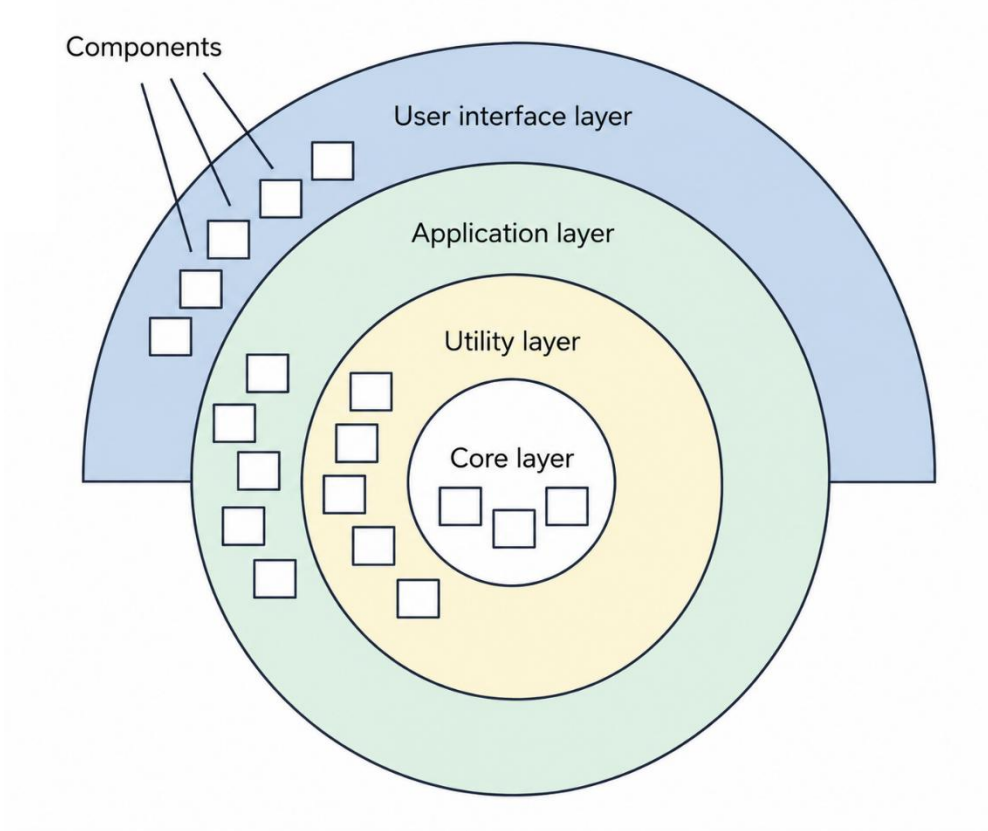




Client-Server



Pipes-and-Filter




Layering

# Architectural Styles and Strategies

## Combining Architectural Styles

- Actual software architectures rarely based on purely one style
- Architectural styles can be combined in several ways
  - Use different styles at different layers (e.g., overall client-server architecture with server component decomposed into layers)
  - Use mixture of styles to model different components or types of interaction (e.g., client components interact with one another using publish-subscribe communications
- If architecture is expressed as collection of models, documentation must be created to show relation between models

# Documenting Software Architectures

- System's architecture is vital to overall development and serves as the basis on decisions for:
  - Design
  - Quality assurance
  - Project management
- The SAD serves as the repository for design information and includes:
  - System overview
  - Views
  - Software units
  - Analysis data and results
  - Design rationale
  - Definitions, glossary, acronyms

# Architecture Design – Review

- Design review is an essential part of engineering practice
- SAD quality is evaluated in two ways:
  - **Validation:** making sure the design satisfies all of the customer's requirements.
  - **Verification:** ensuring the design adheres to good design principles.

# Characteristics of Good Design

- Component independence
  - coupling
  - cohesion
- Exception identification and handling
- Fault prevention and tolerance
  - active
  - passive

# Design Principles

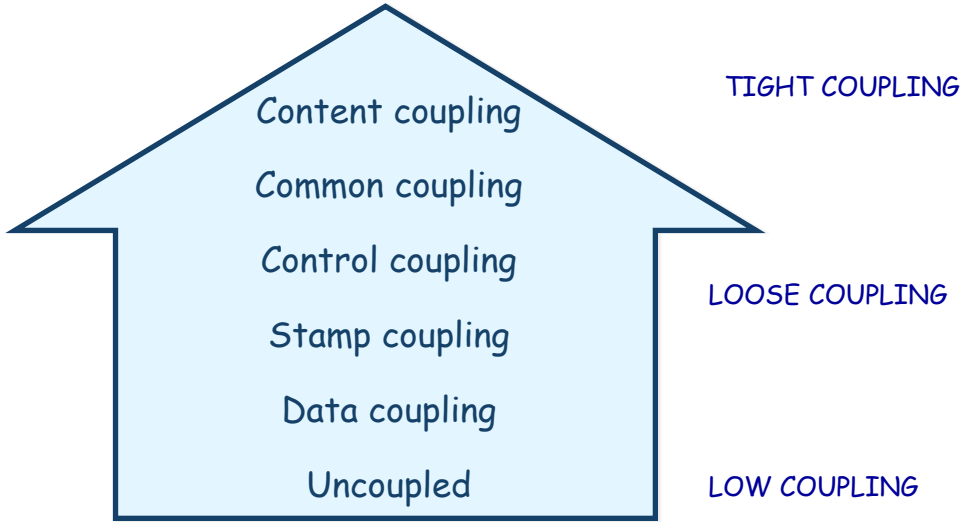
- **Design principles** are guidelines for decomposing a system's required functionality and behavior into modules
- The principles identify the criteria
  - for decomposing a system
  - deciding what information to provide (and what to conceal) in the resulting modules
- Six dominant principles:
  - Modularity
  - Interfaces
  - Information hiding
  - Incremental development
  - Abstraction
  - Generality

See the appendix

# Design Principles

## Modularity - Coupling: Types of Coupling

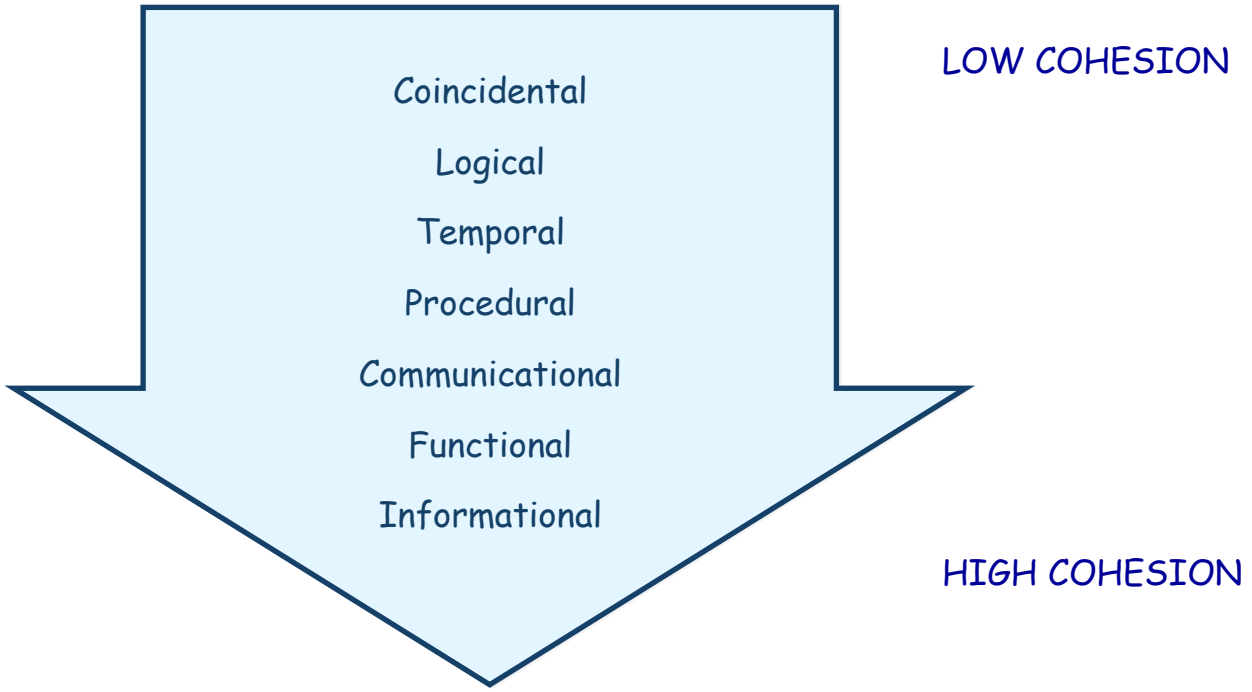
- Content coupling
- Common coupling
- Control coupling
- Stamp coupling
- Data coupling



# Design Principles

## Modularity – Cohesion

- **Cohesion** refers to the dependence within and among a module's internal elements (e.g., data, functions, internal modules)



# Data Design

- Data design specifies the data structures for implementing the software by converting data objects and their relationships identified during the analysis phase.
- Databases

# Data Design

- **Phases of Database Design** (*Rational Database*)
  - Conceptual database design is to build one (or more) **conceptual data models** of the data requirements of the enterprise.
  - Logical database design results in the creation of a **logical data model** of the part of the enterprise that we interested in modeling. The conceptual data model created in the previous phase is refined and mapped onto a logical data model.
  - Physical database design. It's main aim describe how we intend to **physically implement** the logical database design. It producing a description of the implementation of the database on secondary storage; it describes the base relations, file organizations, and indexes used to achieve efficient access to the data, and any associated integrity constraints and security measures.

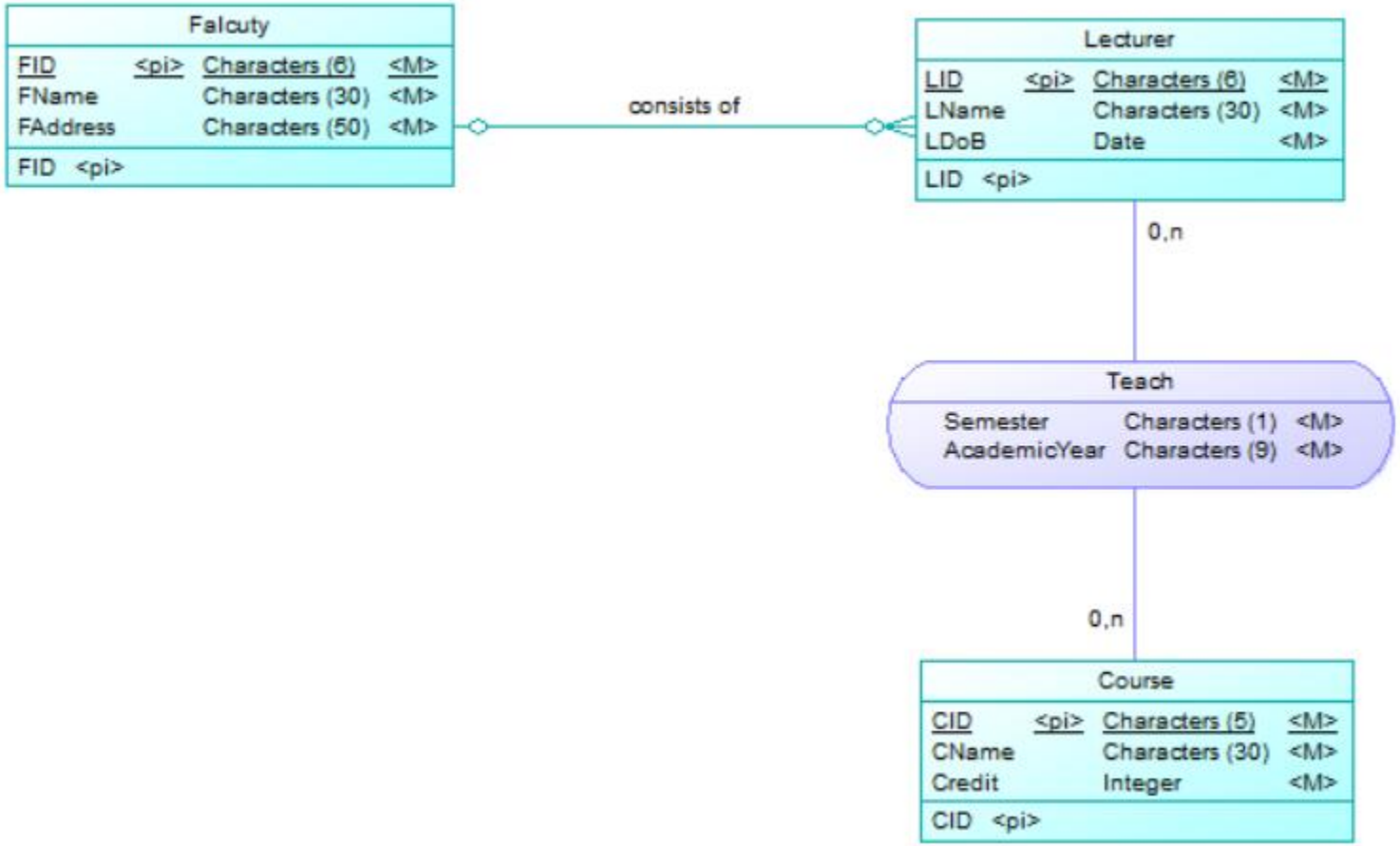
- A conceptual data model comprises: entity types; relationship types; attributes and attribute domains; primary keys and alternate keys; integrity constraints.
- *Build Conceptual Data Model*
  - Identify entity types
  - Identify relationship types
  - Identify and associate attributes with entity or relationship types
  - Determine attribute domains
  - Determine candidate, primary, alternate key attributes
  - Consider use of enhanced modeling concepts (optional step)
  - Check model for redundancy
  - Validate conceptual data model against user transactions
  - Review conceptual data model with user

# Data Design

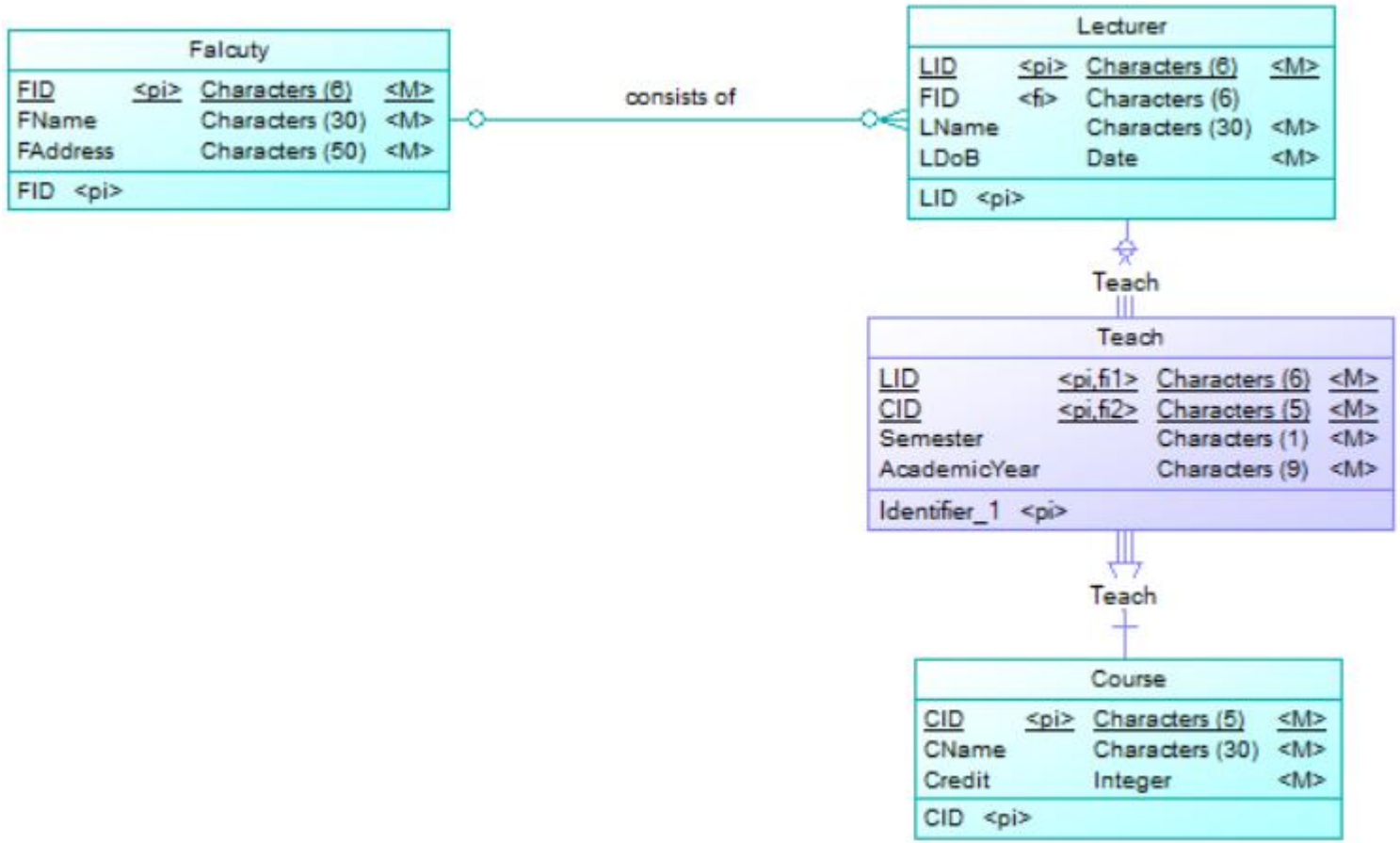
- Build *logical data model*: to translate the conceptual data model into a logical data model and then to validate this model to check that it is structurally correct and able to support the required transactions.
  - Derive relations for logical data model
  - Validate relations using normalization
  - Validate relations against user transactions
  - Check integrity constraints
  - Review logical data model with user
  - Merge logical data models into global data model (optional step)
  - Check for future growth

- The steps of the physical database design methodology
  - Translate logical data model for target DBMS
    - Design base relations
    - Design representation of derived data
    - Design general constraints
  - Design file organizations and indexes
    - Analyze transactions
    - Choose file organizations
    - Choose indexes
    - Estimate disk space requirements
  - Design user views
  - Design security mechanisms
  - Consider the introduction of controlled redundancy
  - Monitor and tune the operational system

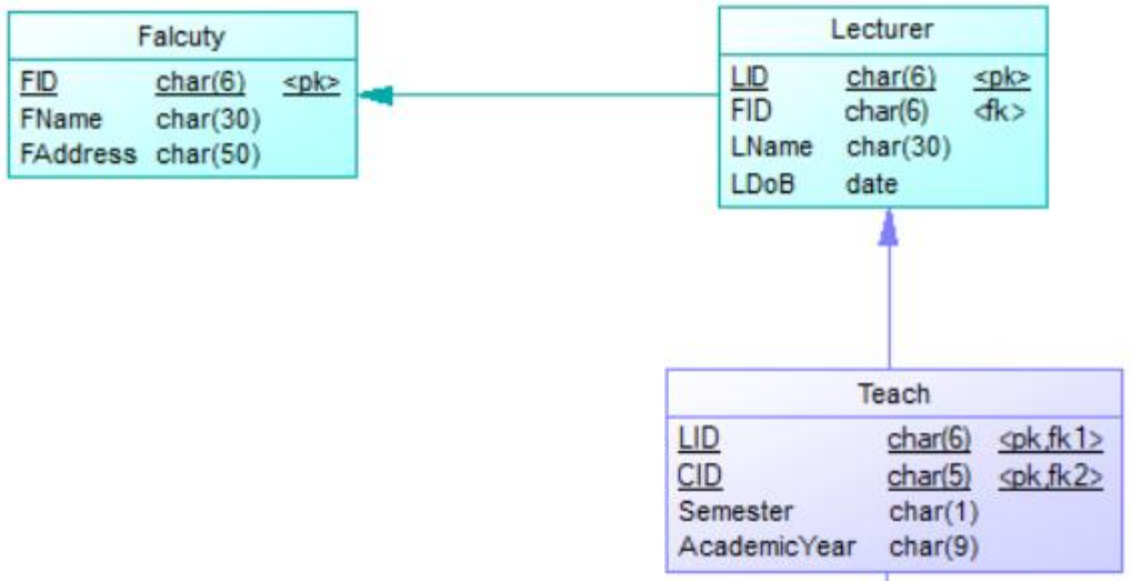
# Data Design – Examples



# Data Design – Examples



# Data Design – Examples

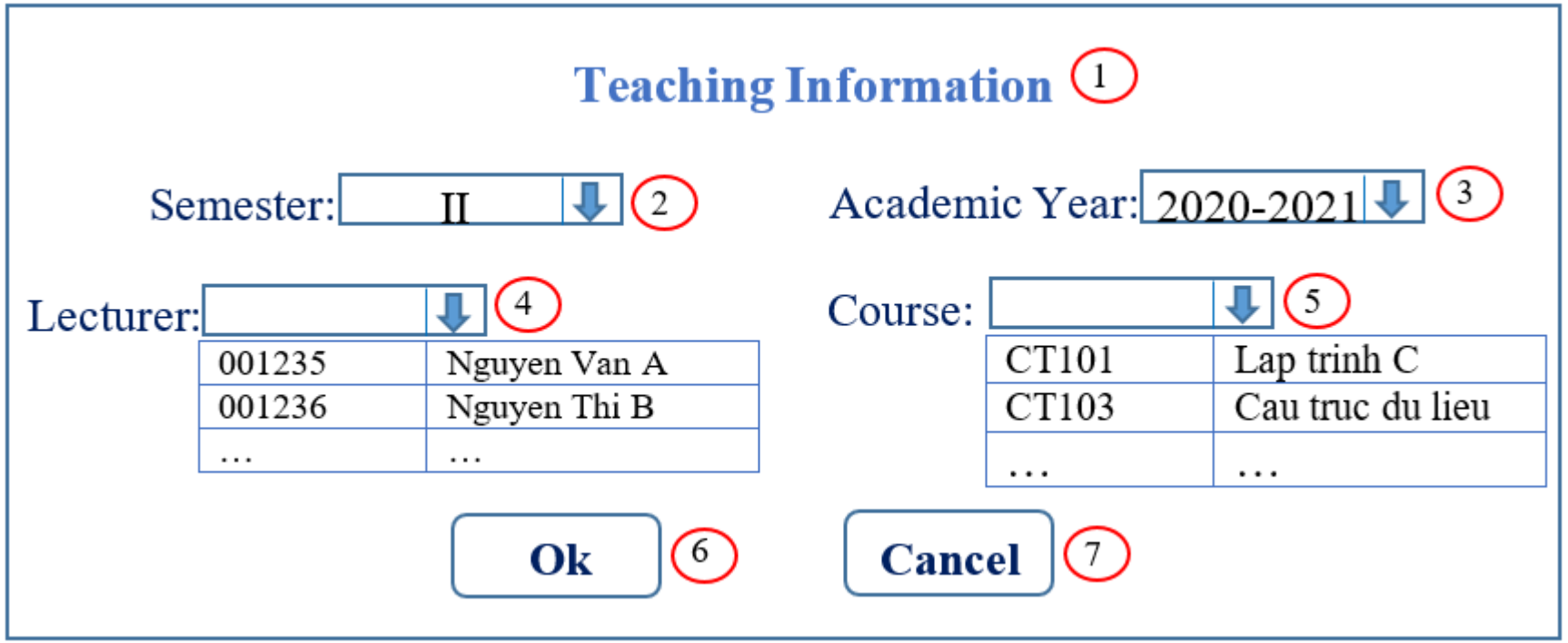



| Name | Code | Data Type | Length | Precision | P | F | M |
|---|---|---|---|---|---|---|---|
| LID | LID | char(6) | 6 | | ☑ | ☑ | ☑ |
| CID | CID | char(5) | 5 | | ☑ | ☑ | ☑ |
| Semester | SEMESTER | char(1) | 1 | | ☐ | ☐ | ☑ |
| AcademicYear | ACADEMICYEA | char(9) | 9 | | ☐ | ☐ | ☑ |

# Interface Design

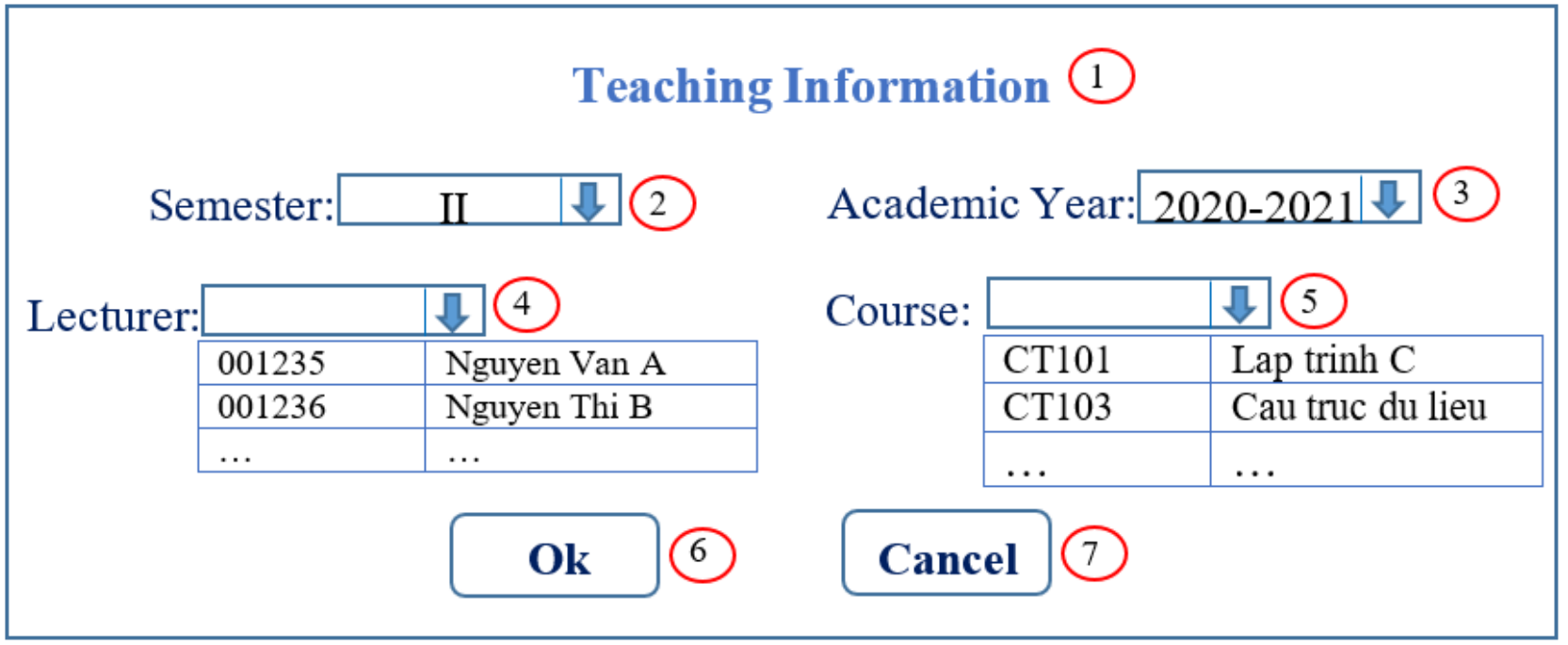
- API Design (System ↔ System)
- UI/UX Design (Human ↔ System)
  - Input forms
  - Reports and other outputs the system must generate

# Interface Design – Examples



# Interface Design – Examples

| No. | Controller type | Default value | Note |
|---|---|---|---|
| 1 | Form | | |
| 2 | Combobox | Current semester | Set: current semester. Values of "Semester" is I, II, III. |
| 3 | Combobox | Current academic year | Set: current academic year. Values of "Academic year - ay" is 2 ay before and after the current ay. |
| 4 | Combobox | | Display LID, LName in table "Lecturer"; select LID |
| 5 | Combobox | | Display CID, CName in table "Course"; select CID |
| 6 | Button | | |
| 7 | Button | | Refresh all comboboxes to the default values. |

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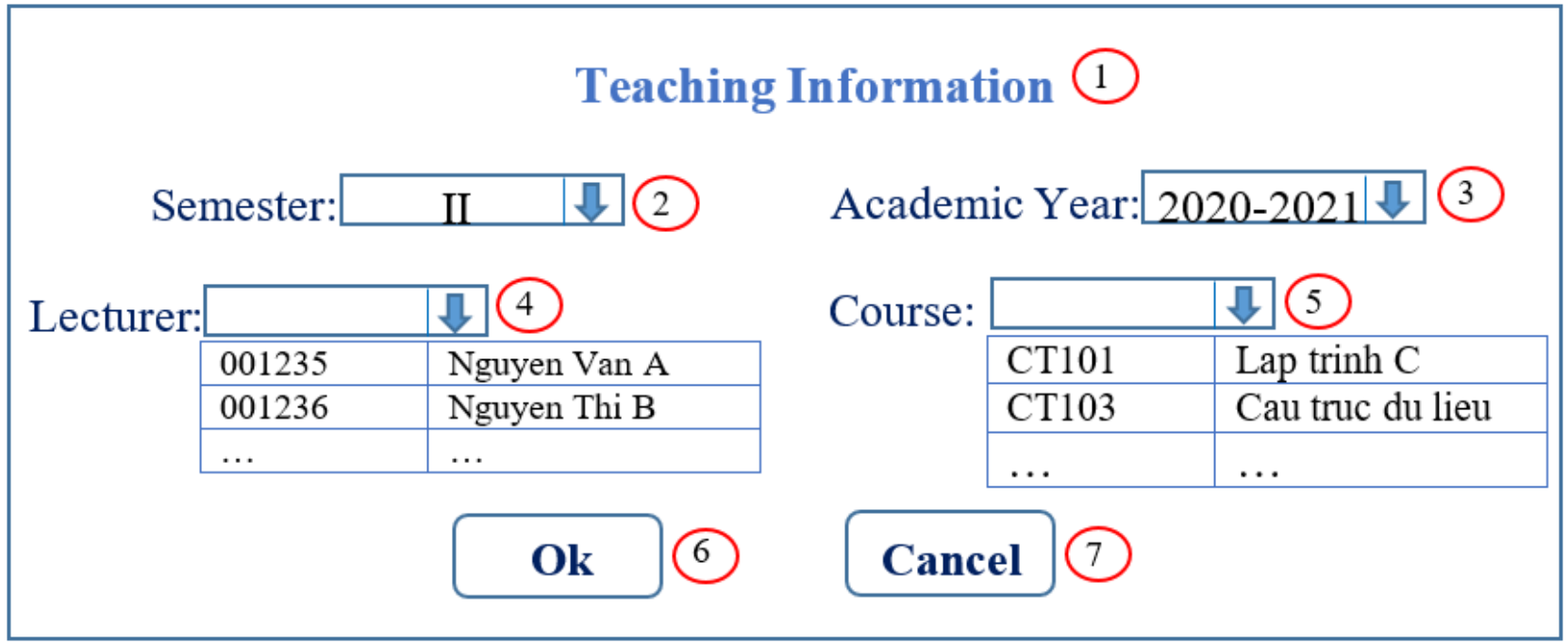
# Interface Design – Examples




| No. | Table name / Data structure | Method | | | |
|---|---|---|---|---|---|
| | | Add | Modify | Delete | Query |
| 1 | Teach | x | | | |
| 2 | Lecturer | | | | x |
| 3 | Course | | | | x |

# Interface Design

- The key issues to be considered in interface design:
  - Cultural issues (nationality, ethnicity, gender, occupation, age, region).
  - Interests of users.
- Notes in the interface design:
  - There should be consistency between the interface (menus, commands, display ...).
  - The names of labels are short and easy to remember.
  - Optimizing in dialog box and the mouse movement.

# Interface Design

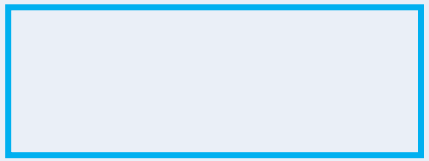
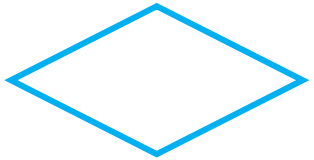
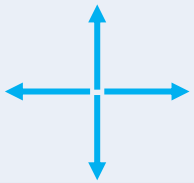
- Notes in the interface design
  - Restricting the direct input data, if possible,  users can choose the available data on the combo box
  - Requesting the confirmation of destructive actions (e.g. erase)
  - Accepting errors from users
  - Should provide feedback to users
  - Creating meaningful error messages
  - Providing the help

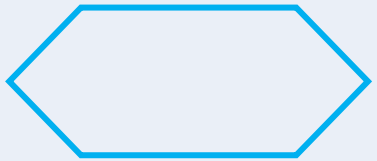
# Detailed Design

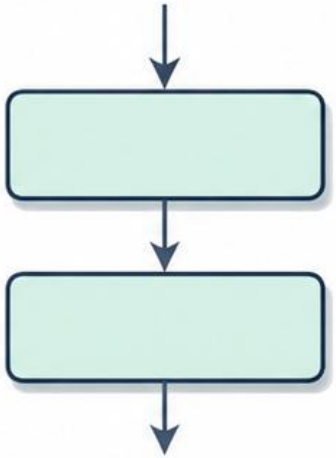
- Structural Design
  - Define classes, attributes, and methods (Class diagram)
- Behavioral Design
  - How objects interact in sequence over time (Sequence diagram)
  - Algorithm flow: branching, loops, and error handling inside a complex function (Flowchart / UML Activity diagram)
- States: managing the lifecycle of an entity (State diagram)

# Graphical representation of an algorithm

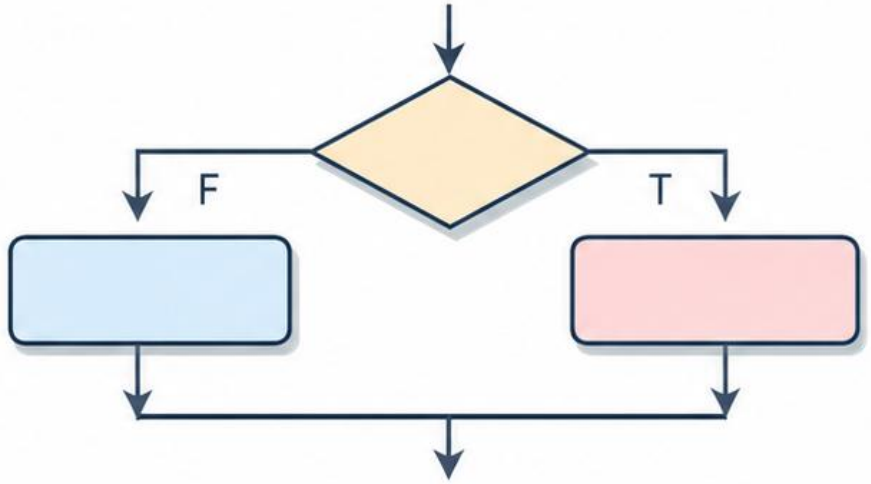
- Algorithm is an ordered set of unambiguous steps that produces a desired output from a given input and terminates in a finite time.
- Presentation methods
  - Flowchart/ UML activity diagram.
  - Pseudocode text
- A flowchart is a graphical representation of an algorithm.

| Symbol Name | Symbol | Function |
|---|---|---|
| Oval |  | Used to represent start and end of flowchart |
| Parallelogram |  | Used for input and output operation |
| Rectangle |  | Processing: Used for arithmetic operations and data manipulations |
| Diamond |  | Decision making: Used to represent operations with two/three alternatives (true/false, etc.) |
| Arrows |  | Flow line: Used to indicate the flow of logic by connecting symbols |
| Circle |  | Page Connector |

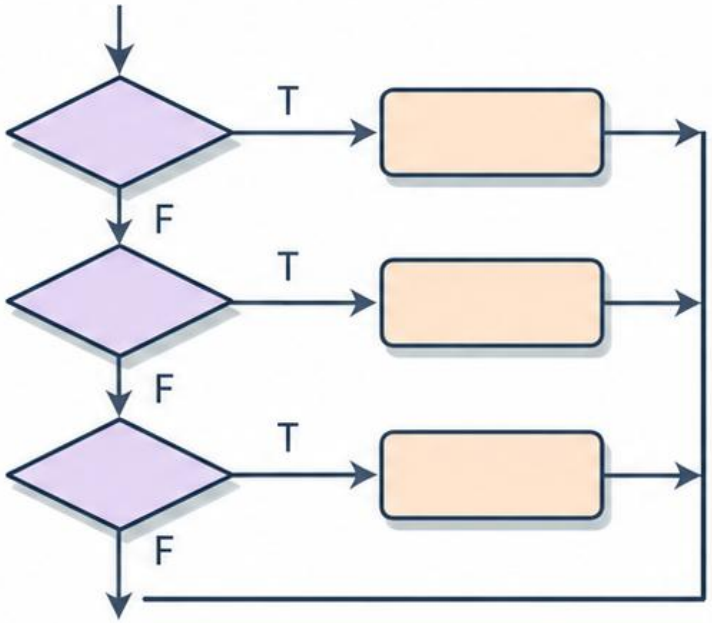
| Symbol Name | | Description |
| --- | --- | --- |
| Off Page Connector |  | |
| Predefined Process/ /Function |  | Used to represent a group of statements performing one processing task. |
| Preprocessor |  | |



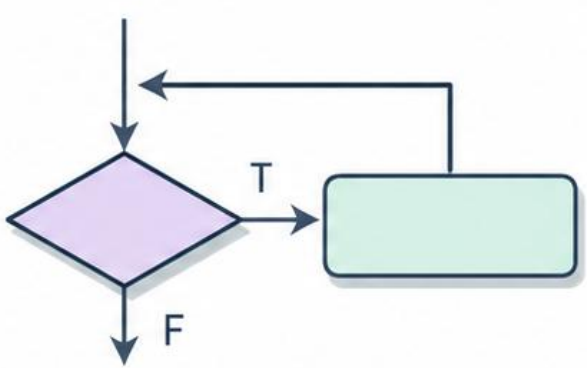
Sequence



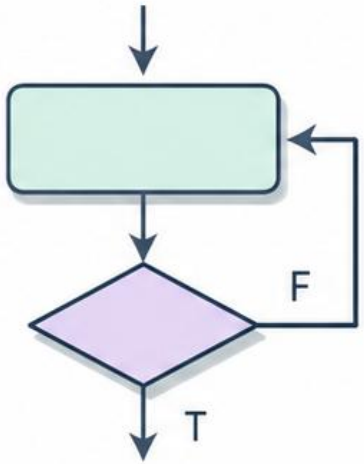

If-then-else




Switch case




Do while



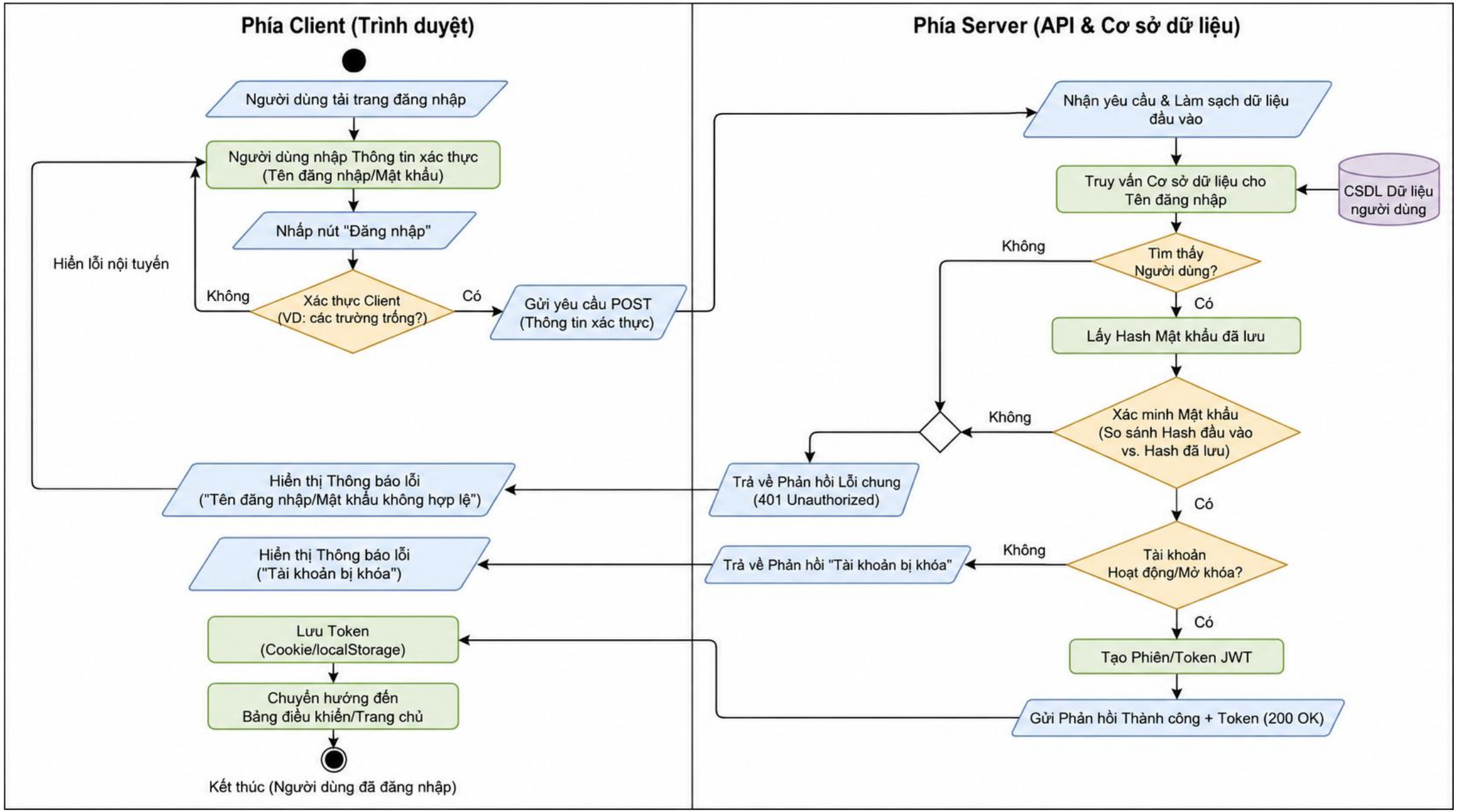

Repeat until

Repetition

# Flowchart – Examples


**Hình 1: Giao diện Đăng Nhập**

# Flowchart – Examples



# Q&A

# Design Principles

- **Design principles** are guidelines for decomposing a system's required functionality and behavior into modules
- The principles identify the criteria
  - for decomposing a system
  - deciding what information to provide (and what to conceal) in the resulting modules
- Six dominant principles:
  - Modularity
  - Interfaces
  - Information hiding
  - Incremental development
  - Abstraction
  - Generality

# Design Principles

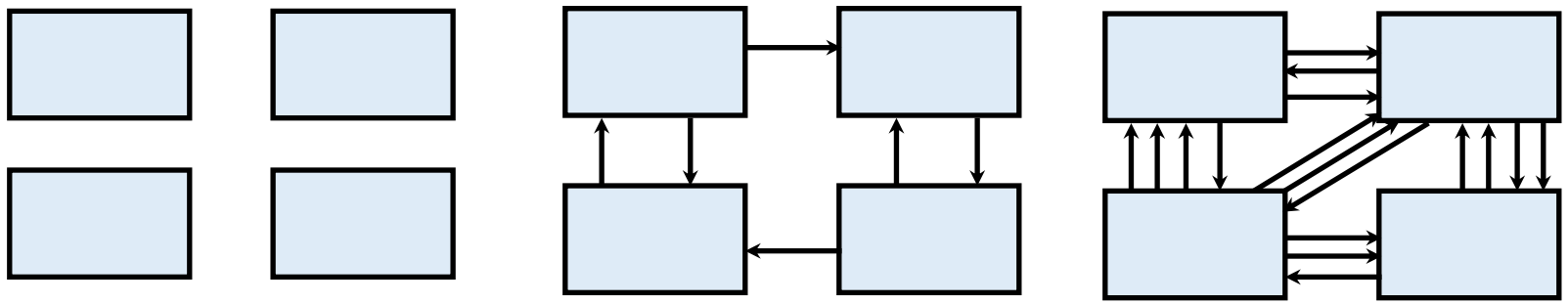
## Modularity

- **Modularity** is the principle of keeping separate the various unrelated aspects of a system, so that each aspect can be studied in isolation (also called separation of concerns)
- If the principle is applied well, each resulting module will have a single purpose and will be relatively independent of the others
  - each module will be easy to understand and develop
  - easier to locate faults (because there are fewer suspect modules per fault)
  - easier to change the system (because a change to one module affects relatively few other modules
- To determine how well a design separates concerns, we use two concepts that measure module independence: coupling and cohesion

# Design Principles

## Modularity - Coupling

- Two modules are **tightly coupled** when they depend a great deal on each other
- **Loosely coupled** modules have some dependence, but their interconnections are weak
- **Uncoupled** modules have no interconnections at all; they are completely unrelated



Uncoupled - no dependencies

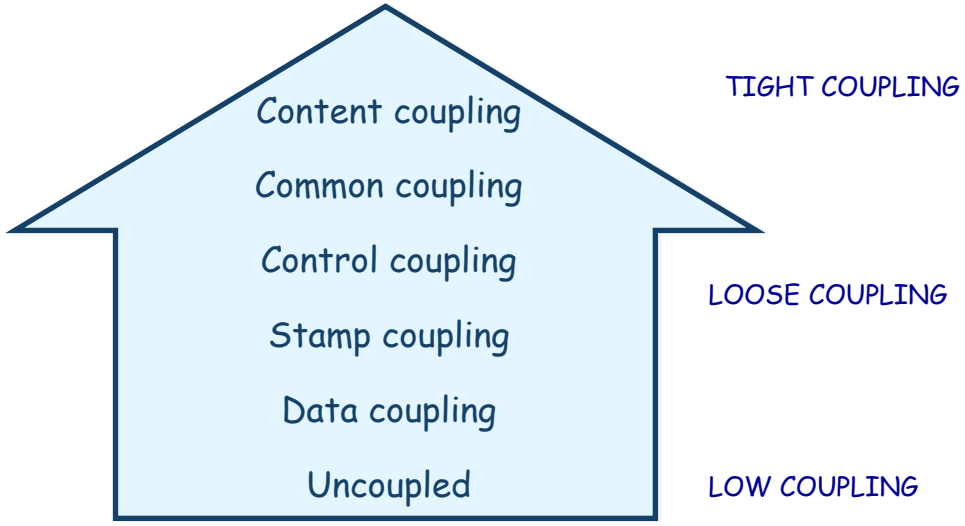
Loosely coupled - some dependencies

Tightly coupled - many dependencies

# Design Principles

## Modularity - Coupling: Types of Coupling

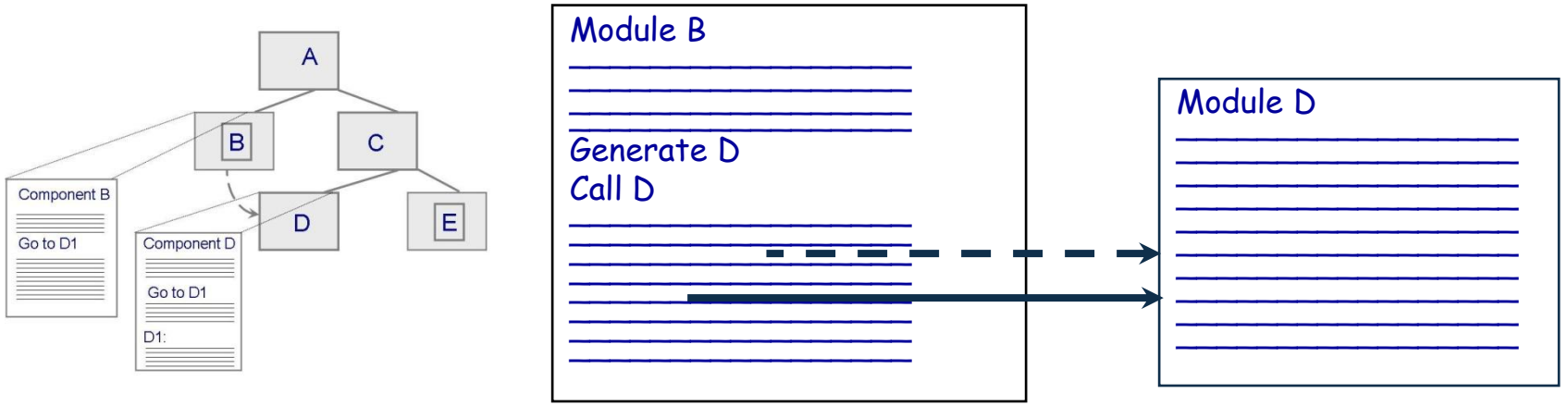
- Content coupling
- Common coupling
- Control coupling
- Stamp coupling
- Data coupling



# Design Principles

## Content Coupling

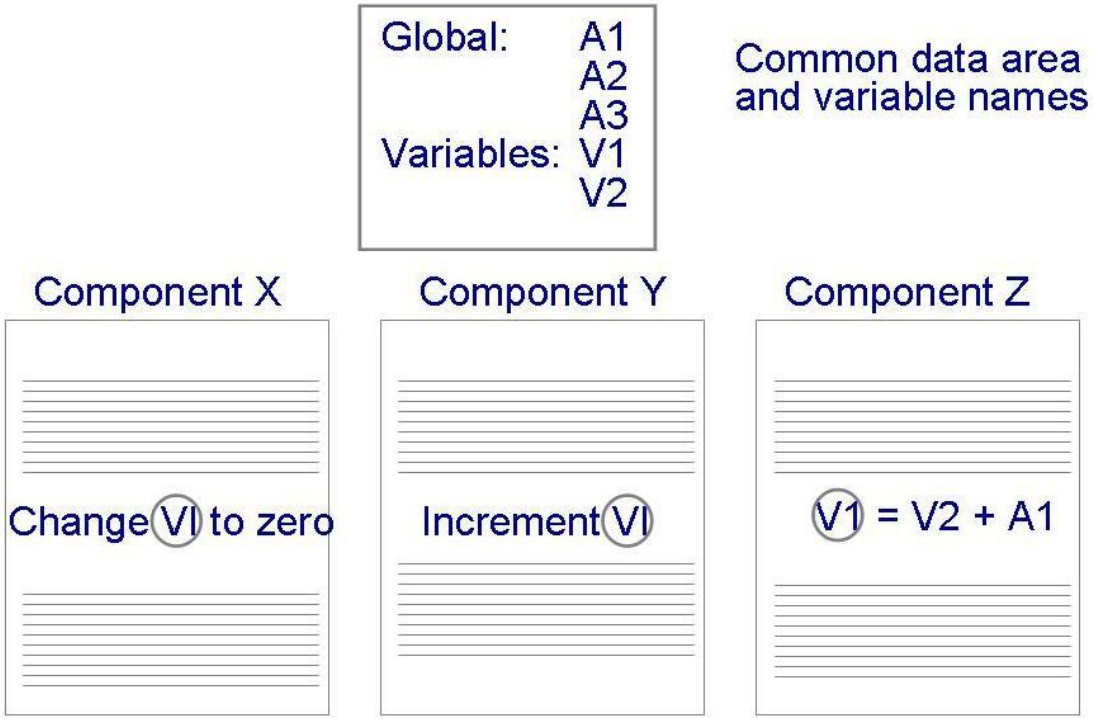
- Occurs when one component modifies an internal data item in another component, or when one component branches into the middle of another component



# Design Principles

## Common Coupling

- Making a change to the common data means tracing back to all components that access those data to evaluate the effect of the change



# Design Principles

## Control Coupling

- When one module passes parameters or a return code to control the behavior of another module
- It is impossible for the controlled module to function without some direction from the controlling module

# Design Principles

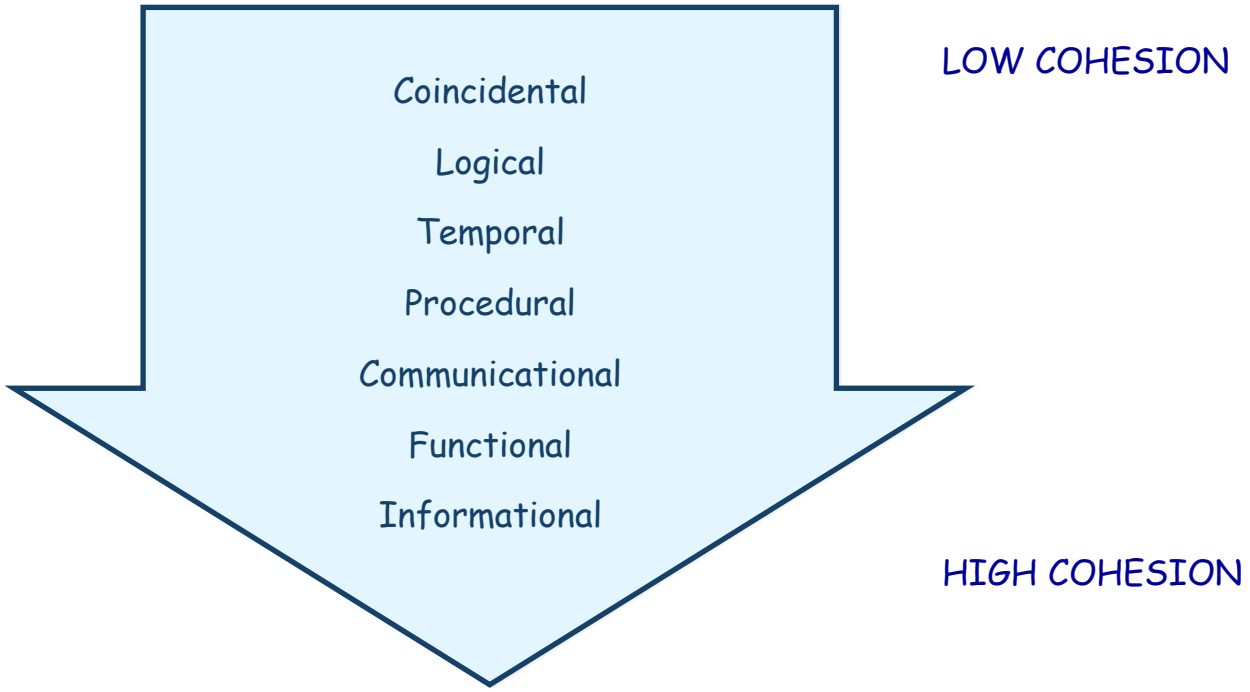
## Stamp and Data Coupling

- **Stamp coupling** occurs when complex data structures are passed between modules
  - Stamp coupling represents a more complex interface between modules, because the modules have to agree on the data's format and organization
- If only data values, and not structured data, are passed, then the modules are connected by **data coupling**
  - Data coupling is simpler and less likely to be affected by changes in data representation

# Design Principles

## Modularity – Cohesion

- **Cohesion** refers to the dependence within and among a module's internal elements (e.g., data, functions, internal modules)



# Design Principles

## Modularity - Cohesion

- Coincidental  (worst degree)
  - Parts are unrelated to one another
- Logical
  - Parts are related only by the logic structure of code
- Temporal
  - Module's data and functions related because they are used at the same time in an execution
- Procedural
  - Similar to temporal, and functions pertain to some related action or purpose

# Design Principles

## Modularity - Cohesion

- Communication
  - Operates on the same data set
- Functional (ideal degree)
  - All elements essential to a single function are contained in one module, and all of the elements are essential to the performance of the function
- Informational
  - Adaption of functional cohesion to data abstraction and object-based design

# Design Principles

## Interfaces

- An **interface** defines what services the software unit provides to the rest of the system, and how other units can access those services
  - For example, the interface to an object is the collection of the object's public operations and the operations' **signatures**, which specify each operation's name, parameters, and possible return values
- An interface must also define what the unit requires, in terms of services or assumptions, for it to work correctly
- A software unit's interface describes what the unit requires of its environment, as well as what it provides to its environment

# Design Principles

## Information Hiding

- **Information hiding** is distinguished by its guidance for decomposing a system:
  - Each software unit encapsulates a separate design decision that could be changed in the future
  - Then the interfaces and interface specifications are used to describe each software unit in terms of its externally visible properties
- Using this principle, modules may exhibit different kinds of cohesion
  - A module that hides a data representation may be informationally cohesive
  - A module that hides an algorithm may be functionally cohesive
- A big advantage of information hiding is that the resulting software units are loosely coupled

# Design Principles

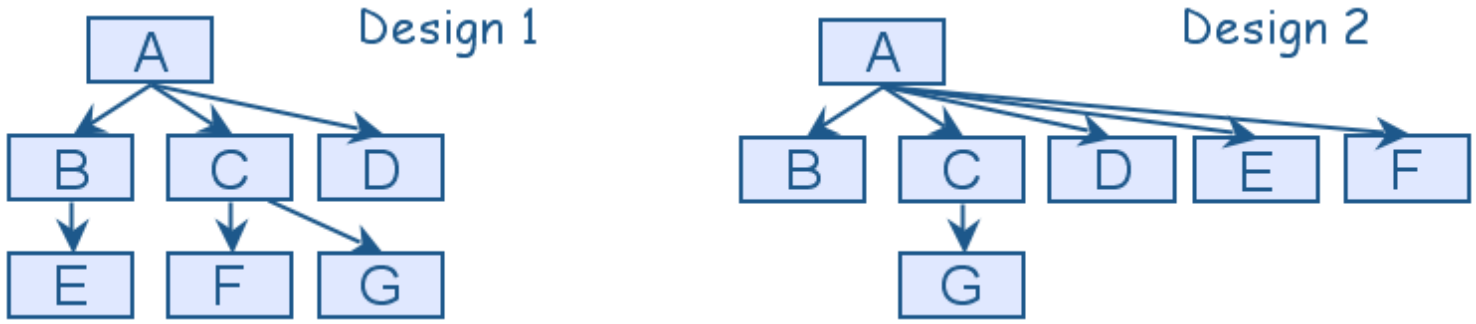
## Incremental Development

- Given a design consisting of software units and their interfaces, we can use the information about the units' dependencies to devise an incremental schedule of development
- Start by mapping out the units' **uses relation**
  - relates each software unit to the other software units on which it depends
- **Uses graphs** can help to identify progressively larger subsets of our system that we can implement and test incrementally

# Design Principles

## Incremental Development

- Uses graphs for two designs
  - **Fan-in** refers to the number of units that use a particular software unit
  - **Fan-out** refers to the number of units used by particular software unit



# Design Principles

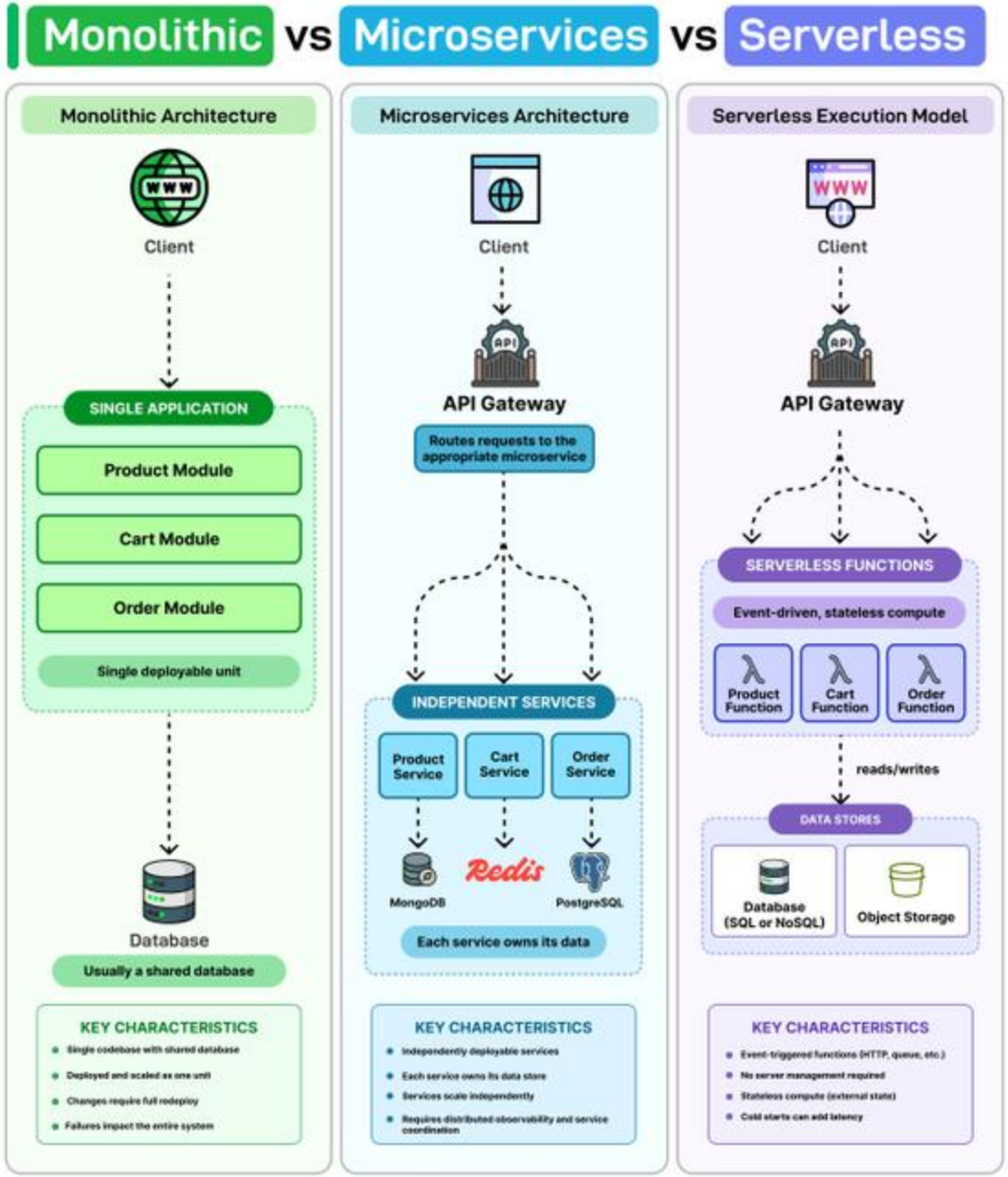
## Abstraction

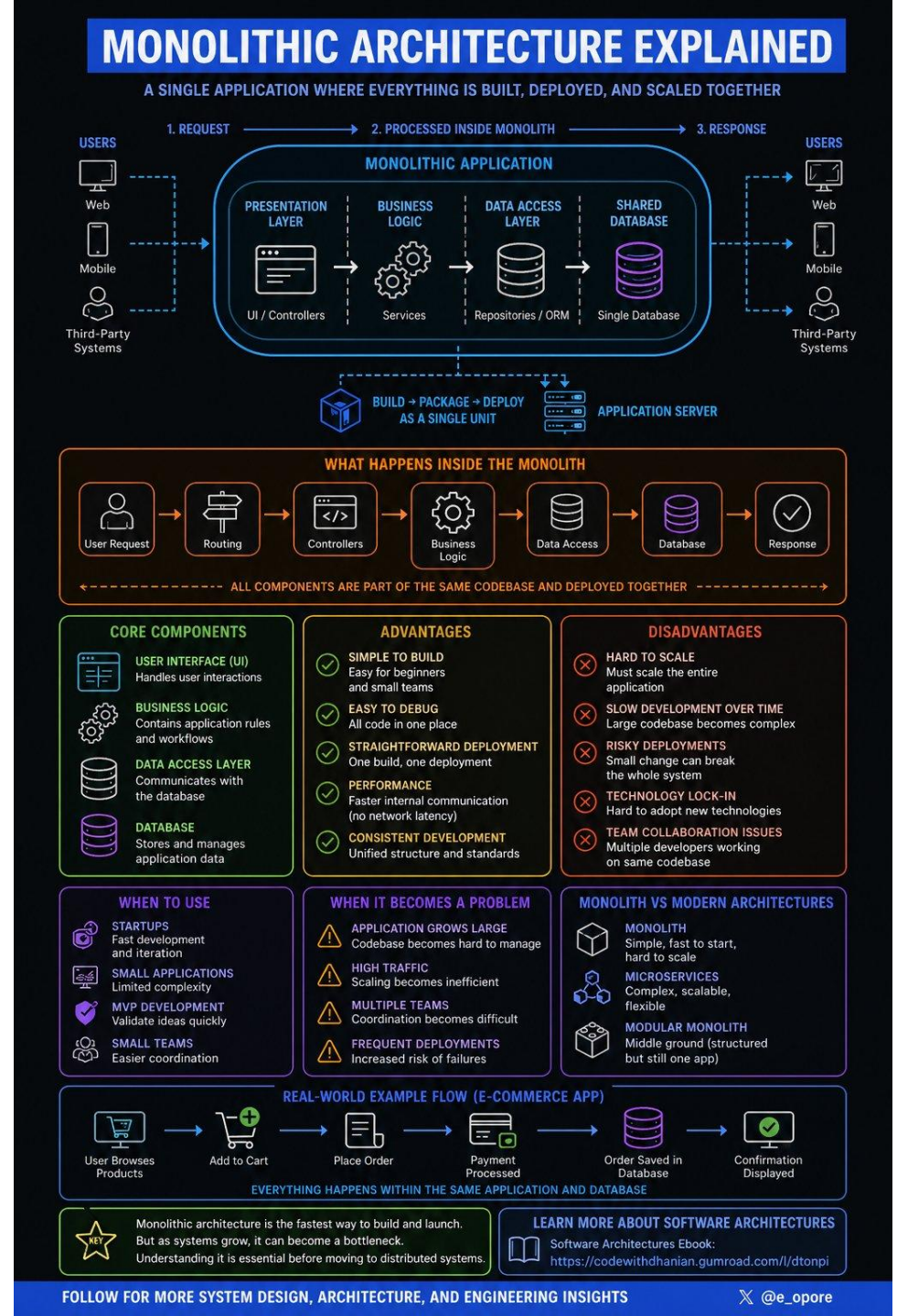
- An **abstraction** is a model or representation that omits some details so that it can focus on other details
- The definition is vague about which details are left out of a model, because different abstractions, built for different purposes, omit different kinds of details

# Design Principles

## Generality

- **Generality** is the design principle that makes a software unit as universally applicable as possible, to increase the chance that it will be useful in some future system
- We make a unit more general by increasing the number of contexts in which can it be used. There are several ways of doing this:
  - Parameterizing context-specific information
  - Removing preconditions
  - Simplifying postconditions


Monolithic vs Microservices vs Serverless
Monolithic Architecture
Client
SINGLE APPLICATION
Product Module
Cart Module
Order Module
Single deployable unit
Database
Usually a shared database
KEY CHARACTERISTICS
Single codebase with shared database
Deployed and scaled as one unit
Changes require full redeploy
Failures impact the entire system
Microservices Architecture
Client
API Gateway
Routes requests to the appropriate microservice
INDEPENDENT SERVICES
Product Service
Cart Service
Order Service
MongoDB
Redis
PostgreSQL
Each service owns its data
KEY CHARACTERISTICS
Independently deployable services
Each service owns its data store
Services scale independently
Requires distributed observability and service coordination
Serverless Execution Model
Client
API Gateway
SERVERLESS FUNCTIONS
Event-driven, stateless compute
Product Function
Cart Function
Order Function
reads/writes
DATA STORES
Database (SQL or NoSQL)
Object Storage
KEY CHARACTERISTICS
Event-triggered functions (HTTP, queue, etc.)
No server management required
Stateless compute (external state)
Cold starts can add latency
ByteByteGo


MONOLITHIC ARCHITECTURE EXPLAINED
A SINGLE APPLICATION WHERE EVERYTHING IS BUILT, DEPLOYED, AND SCALED TOGETHER
1. REQUEST
2. PROCESSED INSIDE MONOLITH
3. RESPONSE
USERS
Web
Mobile
Third-Party Systems
MONOLITHIC APPLICATION
PRESENTATION LAYER
UI / Controllers
BUSINESS LOGIC
Services
DATA ACCESS LAYER
Repositories / ORM
SHARED DATABASE
Single Database
BUILD → PACKAGE → DEPLOY AS A SINGLE UNIT
APPLICATION SERVER
WHAT HAPPENS INSIDE THE MONOLITH
User Request
Routing
Controllers
Business Logic
Data Access
Database
Response
ALL COMPONENTS ARE PART OF THE SAME CODEBASE AND DEPLOYED TOGETHER
CORE COMPONENTS
USER INTERFACE (UI)
Handles user interactions
BUSINESS LOGIC
Contains application rules and workflows
DATA ACCESS LAYER
Communicates with the database
DATABASE
Stores and manages application data
ADVANTAGES
SIMPLE TO BUILD
Easy for beginners and small teams
EASY TO DEBUG
All code in one place
STRAIGHTFORWARD DEPLOYMENT
One build, one deployment
PERFORMANCE
Faster internal communication (no network latency)
CONSISTENT DEVELOPMENT
Unified structure and standards
DISADVANTAGES
HARD TO SCALE
Must scale the entire application
SLOW DEVELOPMENT OVER TIME
Large codebase becomes complex
RISKY DEPLOYMENTS
Small change can break the whole system
TECHNOLOGY LOCK-IN
Hard to adopt new technologies
TEAM COLLABORATION ISSUES
Multiple developers working on same codebase
WHEN TO USE
STARTUPS
Fast development and iteration
SMALL APPLICATIONS
Limited complexity
MVP DEVELOPMENT
Validate ideas quickly
SMALL TEAMS
Easier coordination
WHEN IT BECOMES A PROBLEM
APPLICATION GROWS LARGE
Codebase becomes hard to manage
HIGH TRAFFIC
Scaling becomes inefficient
MULTIPLE TEAMS
Coordination becomes difficult
FREQUENT DEPLOYMENTS
Increased risk of failures
MONOLITH VS MODERN ARCHITECTURES
MONOLITH
Simple, fast to start, hard to scale
MICROSERVICES
Complex, scalable, flexible
MODULAR MONOLITH
Middle ground (structured but still one app)
REAL-WORLD EXAMPLE FLOW (E-COMMERCE APP)
User Browses Products
Add to Cart
Place Order
Payment Processed
Order Saved in Database
Confirmation Displayed
EVERYTHING HAPPENS WITHIN THE SAME APPLICATION AND DATABASE
KEY
Monolithic architecture is the fastest way to build and launch.
But as systems grow, it can become a bottleneck.
Understanding it is essential before moving to distributed systems.
LEARN MORE ABOUT SOFTWARE ARCHITECTURES
Software Architectures Ebook:
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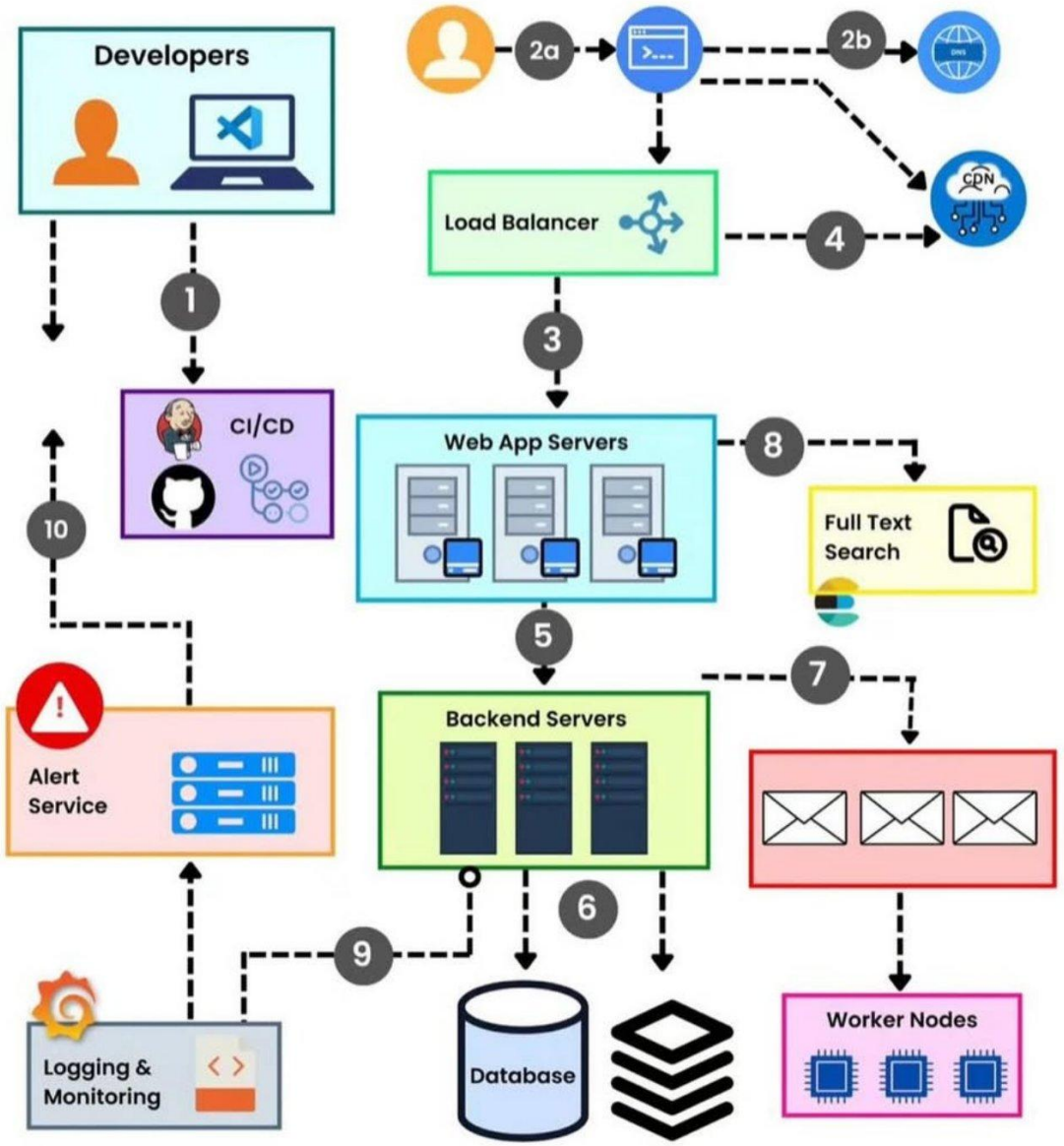
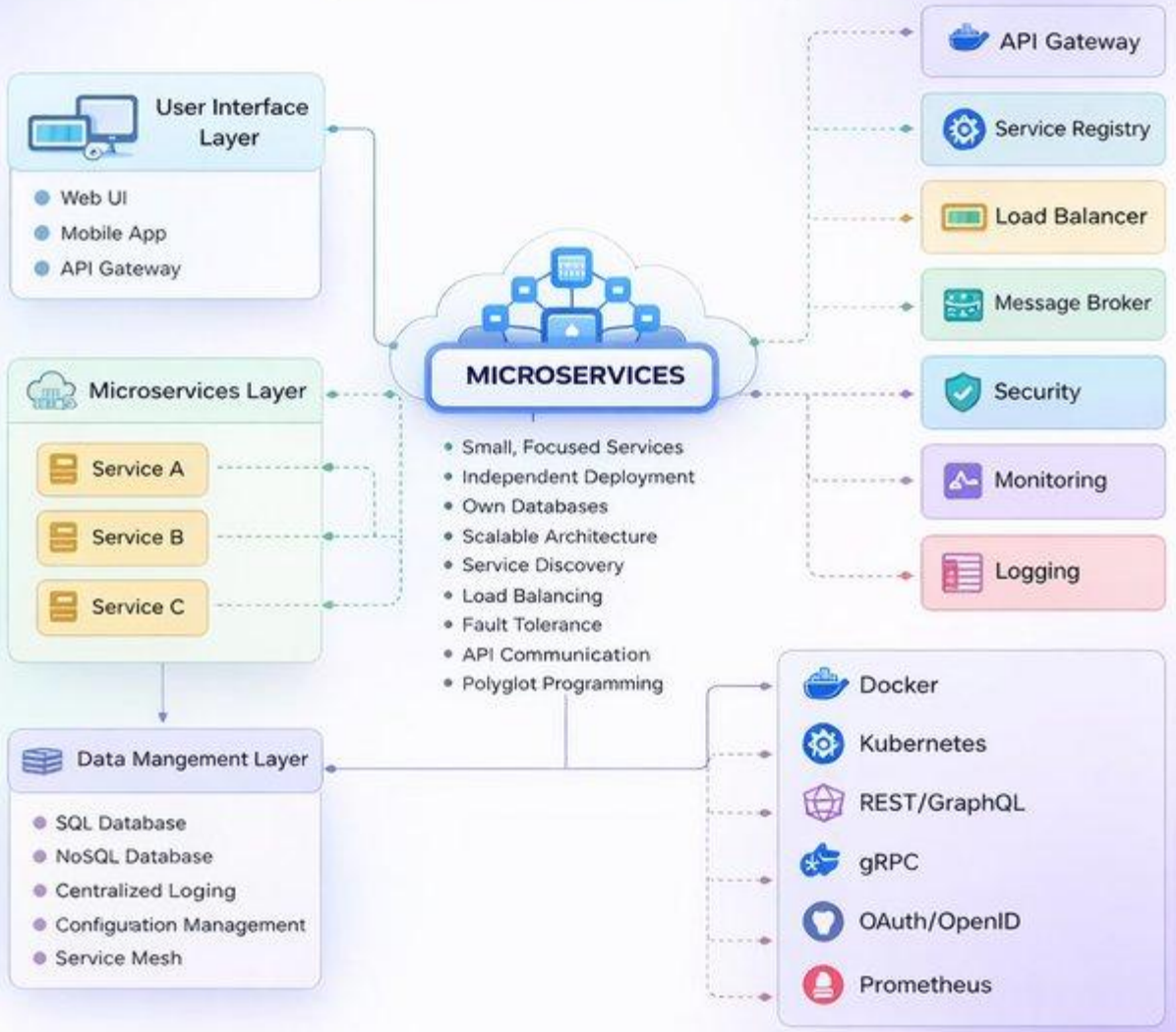
FOLLOW FOR MORE SYSTEM DESIGN, ARCHITECTURE, AND ENGINEERING INSIGHTS
@e_opore

# Typical Architecture of A Web Application



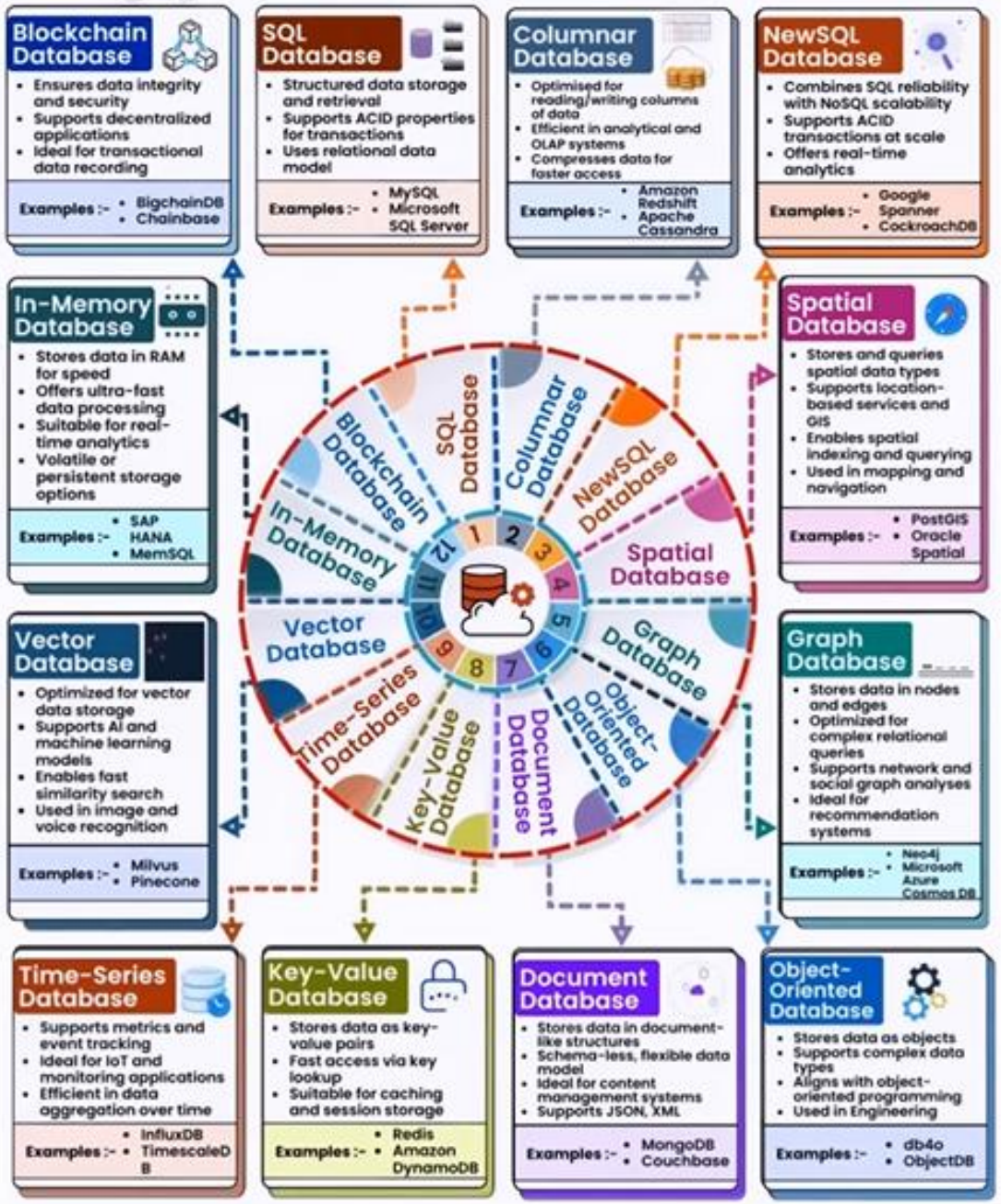

# MICROSERVICES ARCHITECTURE

## System-Level Breakdown



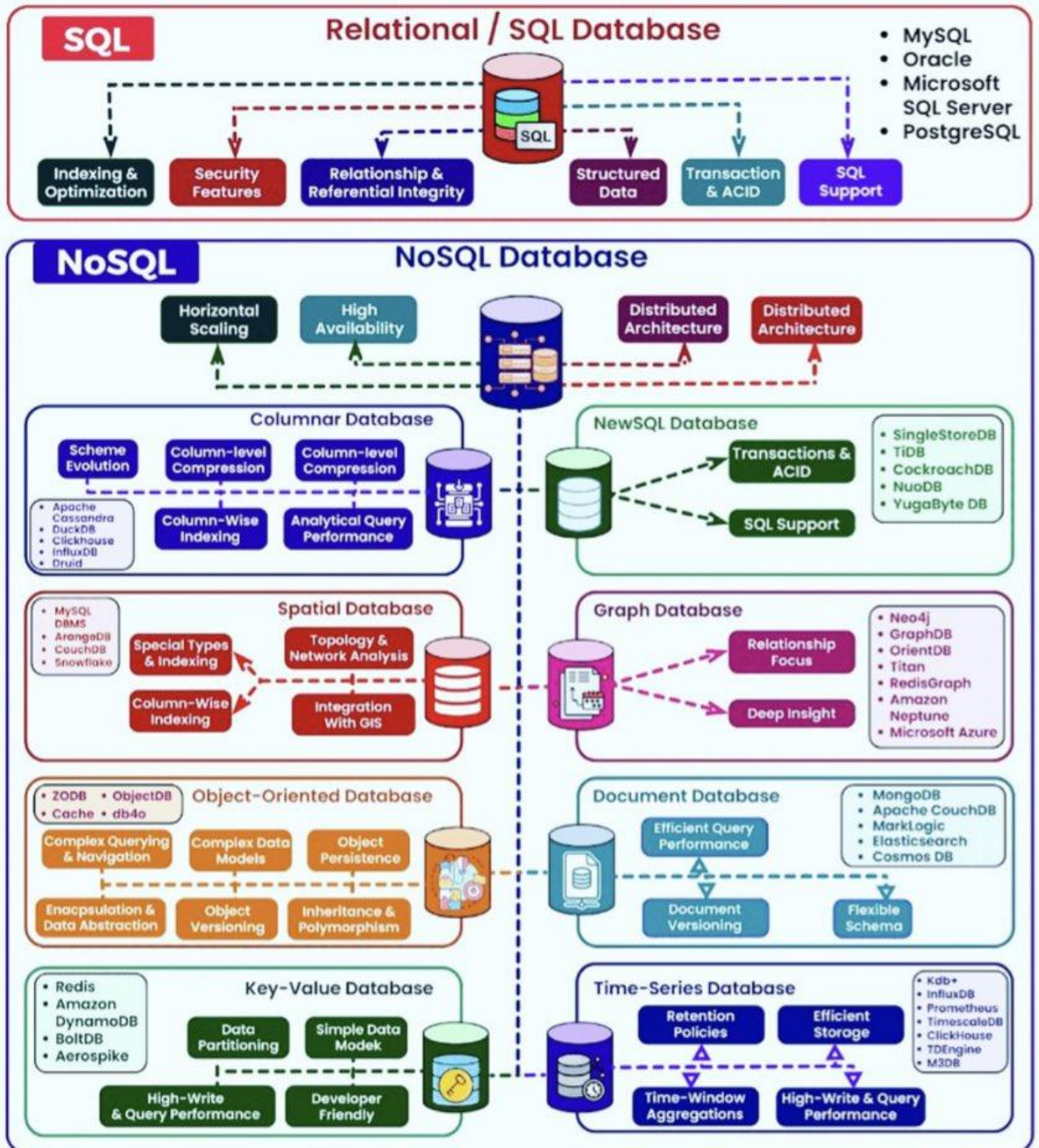

By: Tech Fusionist

# Types of Databases




https://www.linkedin.com/posts/brijpandeyji_understanding-database-options-is-key-for-activity-7160957890193707009-Qiq2?utm_source=share&utm_medium=member_desktop

# Types of Databases




https://www.linkedin.com/pulse/choosing-right-type-db-important-different-types-paresh-nayak-i9shf